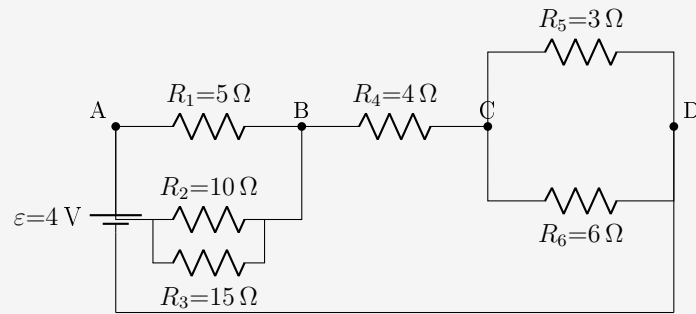


EJERCICIO 1 Dado el circuito: $R_1 = 5 \Omega$, $R_2 = 10 \Omega$, $R_3 = 15 \Omega$, $R_4 = 4 \Omega$, $R_5 = 3 \Omega$, $R_6 = 6 \Omega$, $\varepsilon = 4 \text{ V}$.



a) Resistencia equivalente total R_{eq}

$$\frac{1}{R_{23}} = \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{10} + \frac{1}{15} = \frac{3}{30} + \frac{2}{30} = \frac{5}{30} \Rightarrow R_{23} = 6 \Omega$$

$$\frac{1}{R_{56}} = \frac{1}{R_5} + \frac{1}{R_6} = \frac{1}{3} + \frac{1}{6} = \frac{2}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2} \Rightarrow R_{56} = 2 \Omega$$

Bloque izquierdo (A-B): $R_1 \parallel R_{23}$:

$$\frac{1}{R_{AB}} = \frac{1}{5} + \frac{1}{6} = \frac{6}{30} + \frac{5}{30} = \frac{11}{30} \Rightarrow R_{AB} = \frac{30}{11} \approx 2,73 \Omega$$

$$R_{eq} = R_{AB} + R_4 + R_{56} = 2,73 + 4 + 2 = 8,73 \Omega$$

b) Intensidad total I_T

$$I_T = \frac{\varepsilon}{R_{eq}} = \frac{4}{8,73} \approx 0,458 \text{ A}$$

c) Voltaje entre A y B (V_{AB})

$$V_{AB} = I_T \cdot R_{AB} = 0,458 \times 2,73 \approx 1,25 \text{ V}$$

Intensidades en R_1 y en la rama R_2/R_3 :

$$I_{R1} = \frac{V_{AB}}{R_1} = \frac{1,25}{5} = 0,25 \text{ A} \quad I_{23} = \frac{V_{AB}}{R_{23}} = \frac{1,25}{6} \approx 0,208 \text{ A}$$

$$I_{R2} = \frac{V_{AB}}{R_2} = \frac{1,25}{10} = 0,125 \text{ A} \quad I_{R3} = \frac{V_{AB}}{R_3} = \frac{1,25}{15} \approx 0,083 \text{ A}$$

d) Voltaje entre C y D (V_{CD})

$$V_{CD} = I_T \cdot R_{56} = 0,458 \times 2 = 0,916 \text{ V}$$

$$I_{R5} = \frac{V_{CD}}{R_5} = \frac{0,916}{3} \approx 0,305 \text{ A} \quad I_{R6} = \frac{V_{CD}}{R_6} = \frac{0,916}{6} \approx 0,153 \text{ A}$$

e) Potencia en el bloque A-B (P_{AB})

$$P_{AB} = V_{AB} \cdot I_T = 1,25 \times 0,458 \approx \boxed{0,573 \text{ W}}$$

O bien comprobando por elemento:

$$P_{R1} = \frac{V_{AB}^2}{R_1} = \frac{1,25^2}{5} = 0,313 \text{ W} \quad P_{R2} = \frac{1,25^2}{10} = 0,156 \text{ W} \quad P_{R3} = \frac{1,25^2}{15} = 0,104 \text{ W}$$

f) Potencia en el bloque C-D (P_{CD})

$$P_{CD} = \frac{V_{CD}^2}{R_{56}} = \frac{0,916^2}{2} \approx \boxed{0,419 \text{ W}}$$

$$P_{R5} = \frac{V_{CD}^2}{R_5} = \frac{0,916^2}{3} \approx 0,280 \text{ W} \quad P_{R6} = \frac{0,916^2}{6} \approx 0,140 \text{ W}$$

g) Potencia total (P_{total})

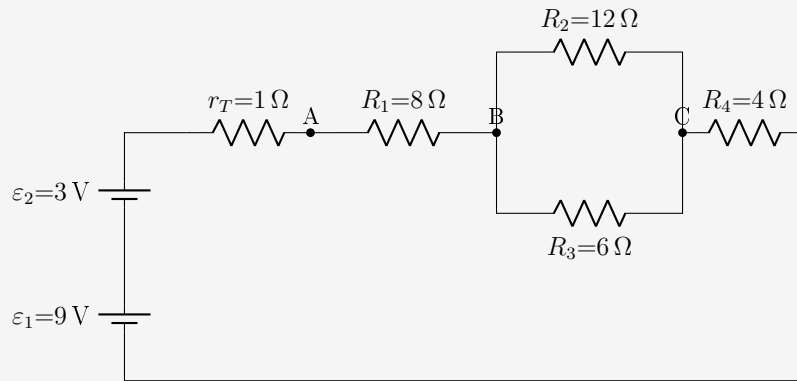
$$P_{total} = \varepsilon \cdot I_T = 4 \times 0,458 \approx \boxed{1,83 \text{ W}}$$

Verificacion: $P_{R1} + P_{R2} + P_{R3} + P_{R4} + P_{R5} + P_{R6}$

$$= 0,313 + 0,156 + 0,104 + (0,458^2 \times 4) + 0,280 + 0,140 = 0,313 + 0,156 + 0,104 + 0,839 + 0,280 + 0,140 \approx 1,83 \text{ W} \checkmark$$

EJERCICIO 2 Dos pilas en serie alimentan el siguiente circuito mixto.

$\varepsilon_1 = 9\text{ V}$, $\varepsilon_2 = 3\text{ V}$, $r_1 = r_2 = 0,5\ \Omega$, $R_1 = 8\ \Omega$, $R_2 = 12\ \Omega$, $R_3 = 6\ \Omega$, $R_4 = 4\ \Omega$.



a) **FEM total y resistencia interna** (pilas en serie, mismo sentido)

$$\varepsilon_T = \varepsilon_1 + \varepsilon_2 = 9 + 3 = \boxed{12\text{ V}} \quad r_T = r_1 + r_2 = 0,5 + 0,5 = 1\ \Omega$$

b) **Resistencia equivalente externa**

$$R_{23} = \frac{R_2 \cdot R_3}{R_2 + R_3} = \frac{12 \times 6}{18} = 4\ \Omega$$

$$R_{ext} = R_1 + R_{23} + R_4 = 8 + 4 + 4 = 16\ \Omega \quad R_{tot} = r_T + R_{ext} = 1 + 16 = \boxed{17\ \Omega}$$

c) **Intensidad total I_T**

$$I_T = \frac{\varepsilon_T}{R_{tot}} = \frac{12}{17} \approx \boxed{0,706\text{ A}}$$

d) **Tension terminal de la bateria**

$$V_{bat} = \varepsilon_T - I_T \cdot r_T = 12 - 0,706 \times 1 \approx \boxed{11,29\text{ V}}$$

e) **Voltaje en cada tramo**

$$V_{R1} = I_T \cdot R_1 = 0,706 \times 8 \approx \mathbf{5,65\text{ V}} \quad V_{BC} = I_T \cdot R_{23} = 0,706 \times 4 \approx \mathbf{2,82\text{ V}} \quad V_{R4} = 0,706 \times 4 \approx \mathbf{2,82\text{ V}}$$

$$\text{Verificacion: } 5,65 + 2,82 + 2,82 \approx 11,29\text{ V} = V_{bat} \checkmark$$

f) **Intensidades en el paralelo B-C**

$$I_{R2} = \frac{V_{BC}}{R_2} = \frac{2,82}{12} \approx 0,235\text{ A} \quad I_{R3} = \frac{V_{BC}}{R_3} = \frac{2,82}{6} \approx 0,47\text{ A}$$

$$0,235 + 0,47 = 0,705\text{ A} \approx I_T \checkmark$$

g) **Potencia en el tramo B-C (P_{BC})**

$$P_{BC} = V_{BC} \cdot I_T = 2,82 \times 0,706 \approx \boxed{1,99\text{ W}}$$

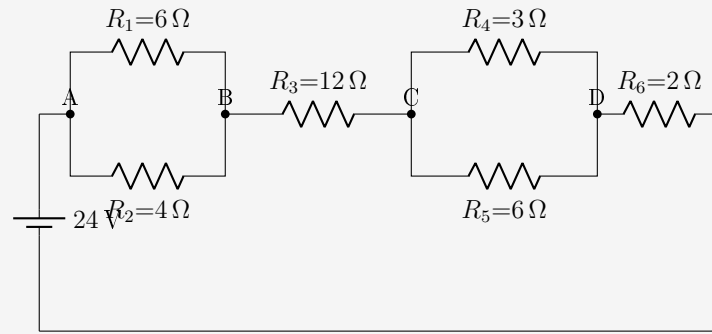
$$P_{R2} = \frac{V_{BC}^2}{R_2} = \frac{2,82^2}{12} \approx 0,663 \text{ W} \quad P_{R3} = \frac{2,82^2}{6} \approx 1,326 \text{ W} \quad \Rightarrow \quad P_{R2} + P_{R3} \approx 1,99 \text{ W} \checkmark$$

h) Potencia total del circuito

$$P_{total} = \varepsilon_T \cdot I_T = 12 \times 0,706 \approx \boxed{8,47 \text{ W}}$$

EJERCICIO 3 Circuito mixto con tres bloques en serie.

$\varepsilon = 24 \text{ V}$, $R_1 = 6 \Omega$, $R_2 = 4 \Omega$, $R_3 = 12 \Omega$, $R_4 = 3 \Omega$, $R_5 = 6 \Omega$, $R_6 = 2 \Omega$.

**a) Resistencia equivalente total**

$$R_{12} = \frac{6 \times 4}{6 + 4} = \frac{24}{10} = 2,4 \Omega \quad R_{45} = \frac{3 \times 6}{3 + 6} = \frac{18}{9} = 2 \Omega$$

$$R_{eq} = R_{12} + R_3 + R_{45} + R_6 = 2,4 + 12 + 2 + 2 = 18,4 \Omega$$

b) Intensidad total

$$I_T = \frac{24}{18,4} \approx \boxed{1,304 \text{ A}}$$

c) Voltaje en el bloque A-B (V_{AB})

$$V_{AB} = I_T \cdot R_{12} = 1,304 \times 2,4 \approx \boxed{3,13 \text{ V}}$$

$$I_{R1} = \frac{3,13}{6} \approx 0,522 \text{ A} \quad I_{R2} = \frac{3,13}{4} \approx 0,783 \text{ A} \quad 0,522 + 0,783 \approx 1,305 \text{ A} \checkmark$$

d) Voltaje en R_3 (tramo B-C)

$$V_{R3} = I_T \cdot R_3 = 1,304 \times 12 \approx \boxed{15,65 \text{ V}}$$

e) Voltaje en el bloque C-D (V_{CD})

$$V_{CD} = I_T \cdot R_{45} = 1,304 \times 2 \approx \boxed{2,61 \text{ V}}$$

$$I_{R4} = \frac{2,61}{3} \approx 0,87 \text{ A} \quad I_{R5} = \frac{2,61}{6} \approx 0,435 \text{ A} \quad 0,87 + 0,435 \approx 1,305 \text{ A} \checkmark$$

f) Potencia en el bloque A-B

$$P_{AB} = V_{AB} \cdot I_T = 3,13 \times 1,304 \approx \boxed{4,08 \text{ W}}$$

$$P_{R1} = \frac{3,13^2}{6} \approx 1,63 \text{ W} \quad P_{R2} = \frac{3,13^2}{4} \approx 2,45 \text{ W} \quad \Rightarrow \quad 1,63 + 2,45 \approx 4,08 \text{ W} \checkmark$$

g) Potencia en el bloque C-D

$$P_{CD} = V_{CD} \cdot I_T = 2,61 \times 1,304 \approx \boxed{3,40 \text{ W}}$$

$$P_{R4} = \frac{2,61^2}{3} \approx 2,27 \text{ W} \quad P_{R5} = \frac{2,61^2}{6} \approx 1,13 \text{ W} \quad \Rightarrow \quad 2,27 + 1,13 \approx 3,40 \text{ W} \checkmark$$

h) Potencia total

$$P_{total} = \varepsilon \cdot I_T = 24 \times 1,304 \approx \boxed{31,3 \text{ W}}$$

Verificacion:

$$P_{AB} + P_{R3} + P_{CD} + P_{R6} = 4,08 + (1,304^2 \times 12) + 3,40 + (1,304^2 \times 2) \approx 4,08 + 20,40 + 3,40 + 3,40 = 31,28 \text{ W} \checkmark$$